

■ ■ NASA Prep Apprenticeship Log — Day 2.1

Theme: Image Import & Visualization Experiments

Today's focus is transitioning from environment verification to practical usage — importing sample images, verifying LightGlue's visual pipeline, and documenting observations.

■ Session Goals

- 1. Verify that image imports and file paths are correctly recognized within the project directory.
- 2. Load sample images (e.g., test1.jpg, test2.jpg) and confirm display using OpenCV or Matplotlib.
- 3. Initialize LightGlue + SuperPoint models for feature extraction and visualization.
- 4. Run and document initial matching results between two images.
- 5. Record observations, errors, and adjustments needed for next session.

■ Commands / Scripts Executed

Use this section to list key commands, scripts, or snippets executed during the session. Example: ``python test_lightglue_import.py``, or `image import test` code.

■ Results & Observations

Describe what occurred during image import or visualization: did images display correctly? Were there path or module errors? Include short notes or screenshots if applicable.

■ Reflection & Next Steps

Summarize key takeaways, patterns recognized, or insights gained about the LightGlue image pipeline. Outline next experiments for upcoming sessions (e.g., refining feature extraction, exploring SuperPoint parameters).

■ ■ Time Tracking

Enter total session time here. ChatGPT will convert it to Tufts' format (2.5 = 2 hours 30 minutes).

"Experimentation turns knowledge into understanding."